CHAPTER 10

"REFLECTION OF LIGHT"

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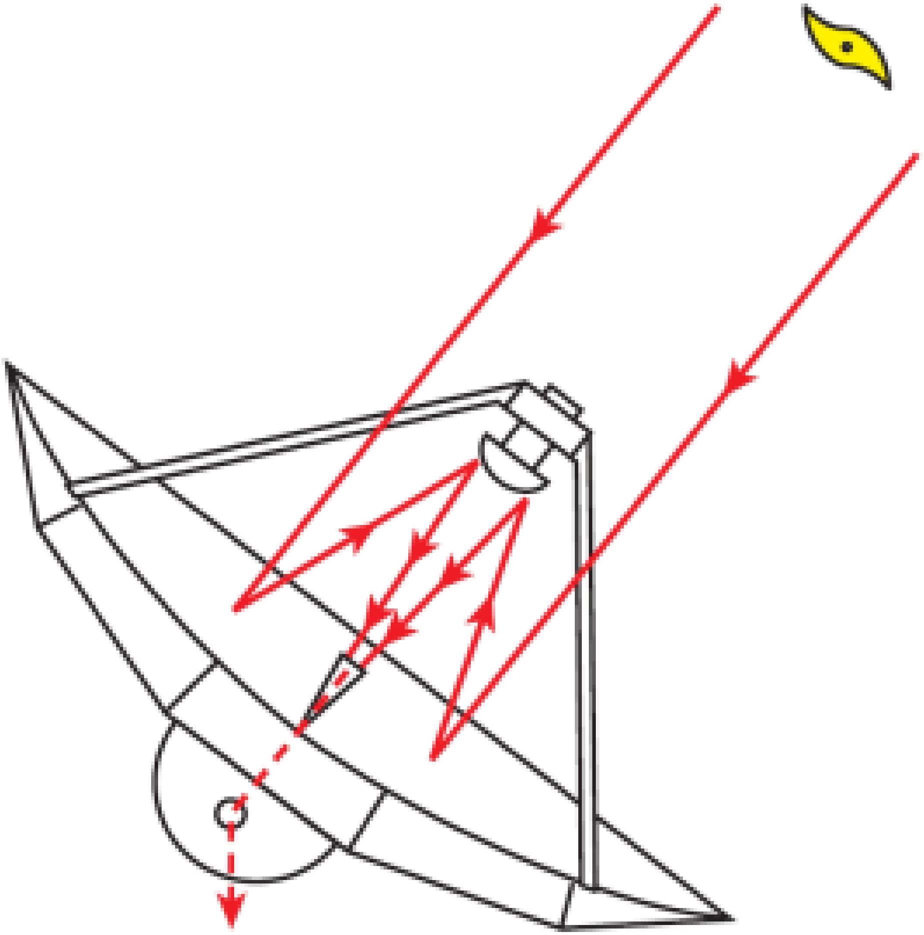

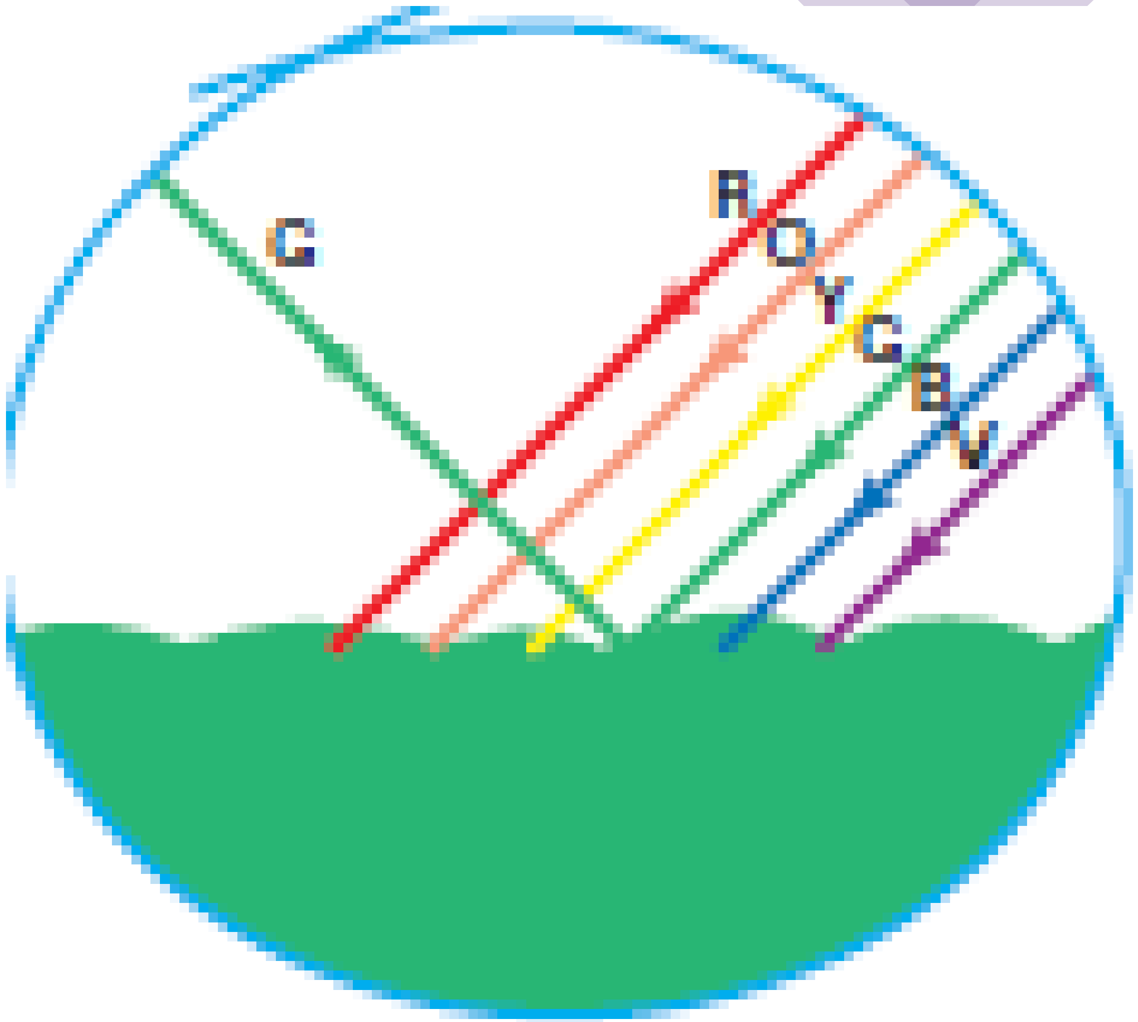
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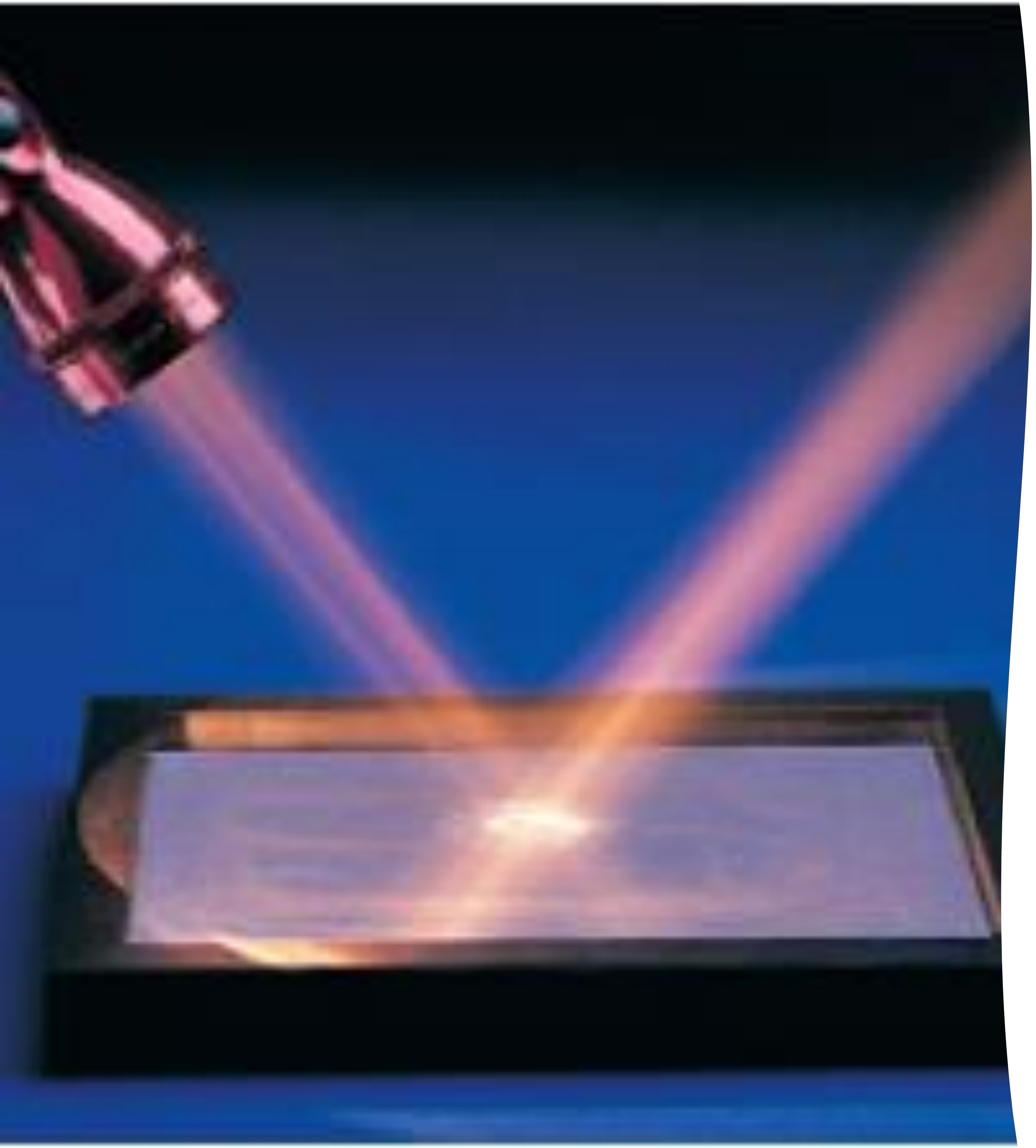
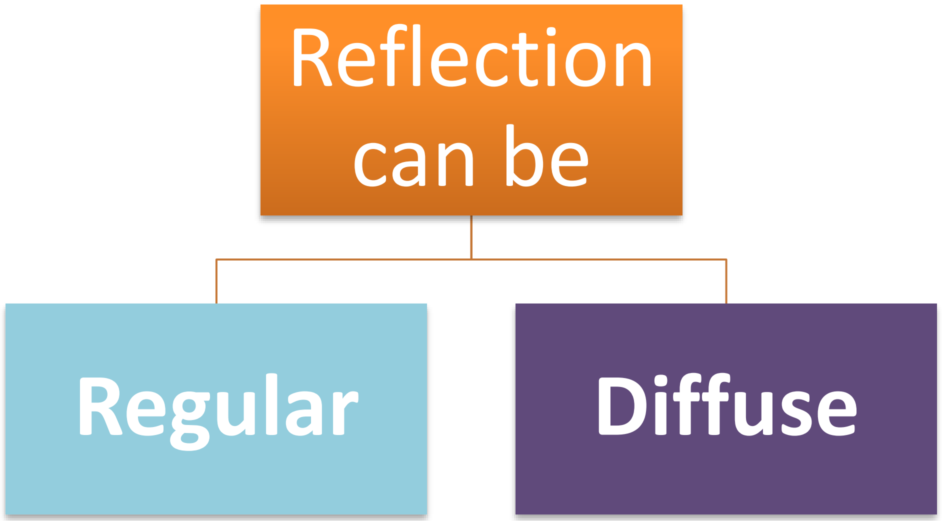
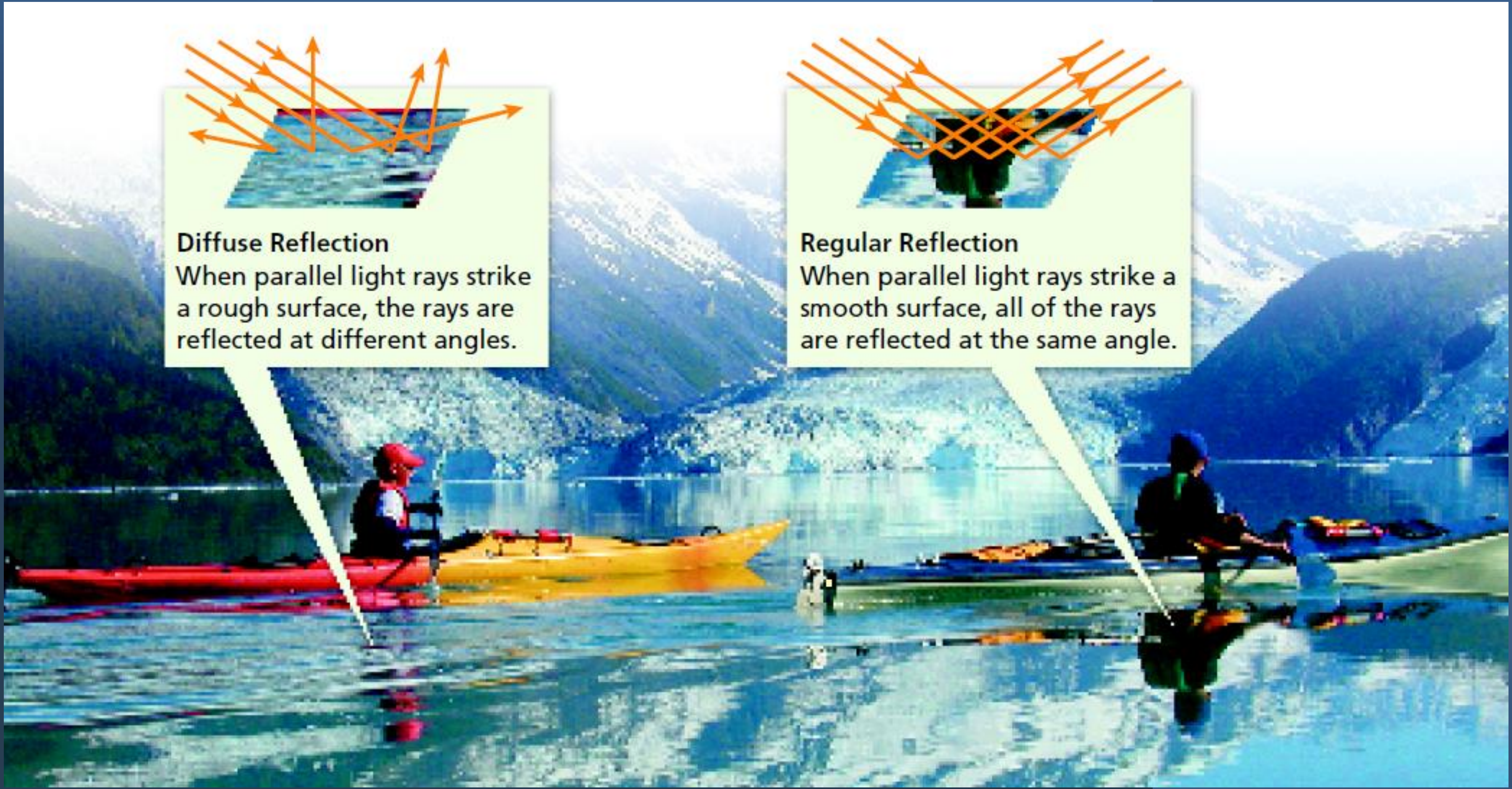


Reflection of Light

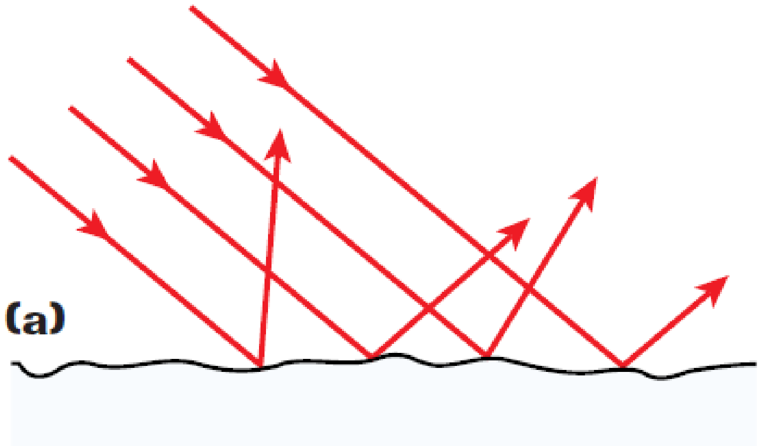
- Like any wave, when like encounters a surface or an obstacle which it cannot pass through, it bounces back.
- This interaction is called reflection.

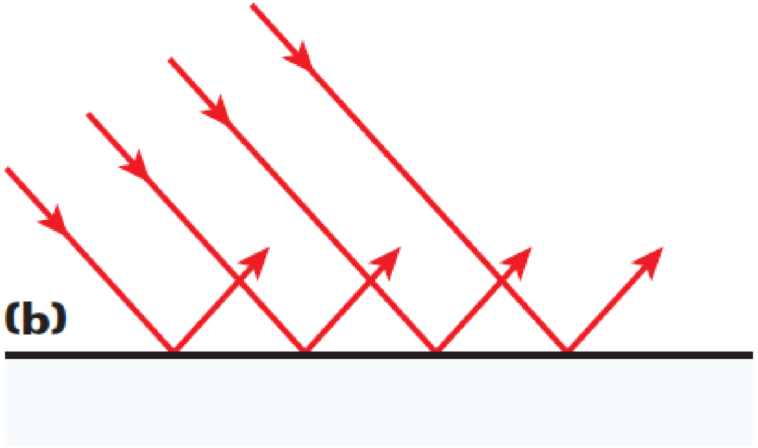
Types of Reflection




Diffuse Reflection
When parallel light rays strike a rough surface, the rays are reflected at different angles.
Regular Reflection
When parallel light rays strike a smooth surface, all of the rays are reflected at the same angle.

The texture of a surface affects how it reflects light





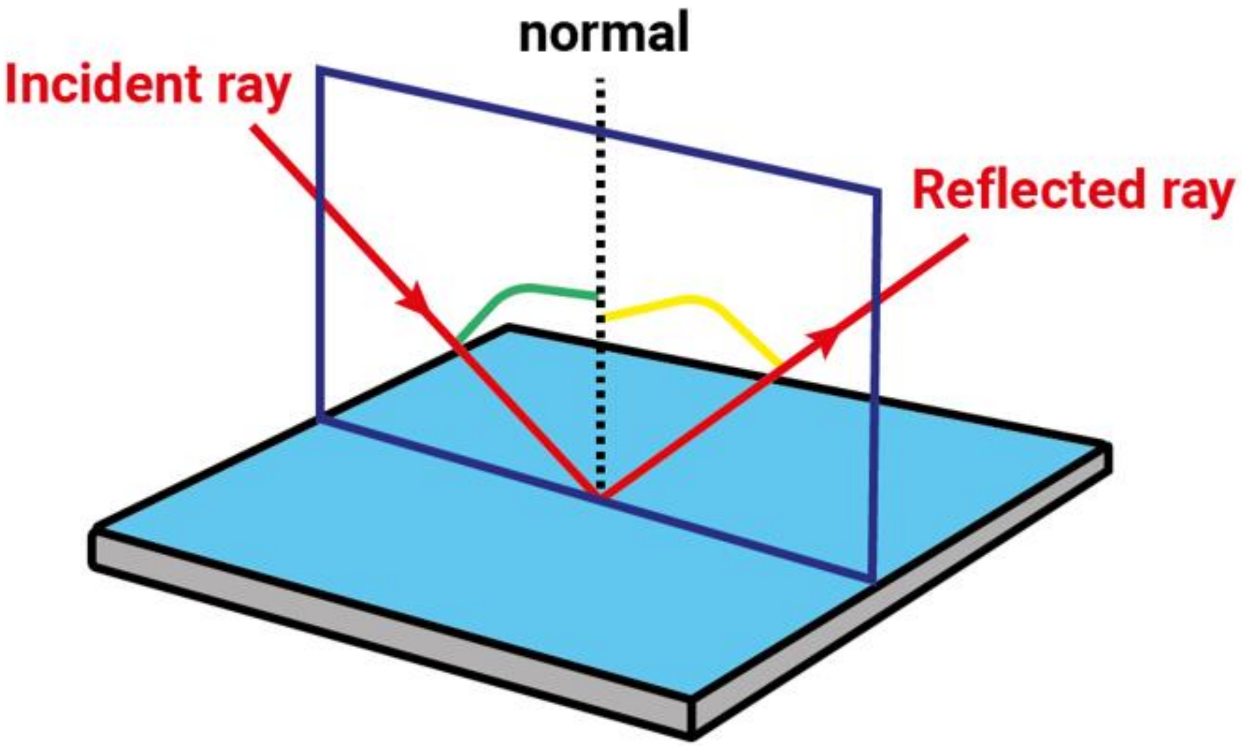
Diffusely reflected light is reflected in many directions

Specularly reflected light is reflected in the same forward direction only

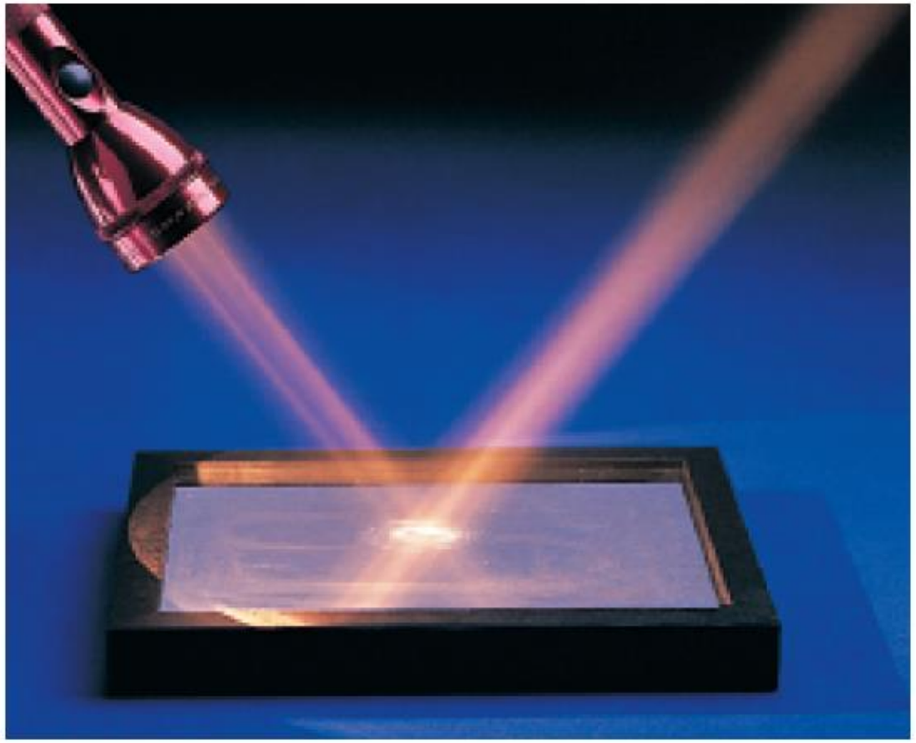
Laws of Reflection

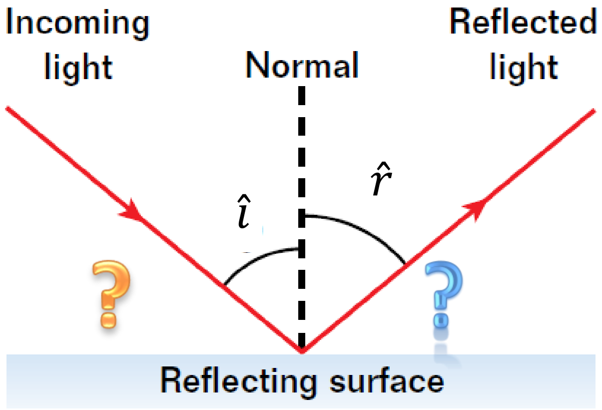
First law:

The **incident ray**, **the reflected ray**, and **the normal** at the point of incidence all lie in the same plane.



Laws of Reflection

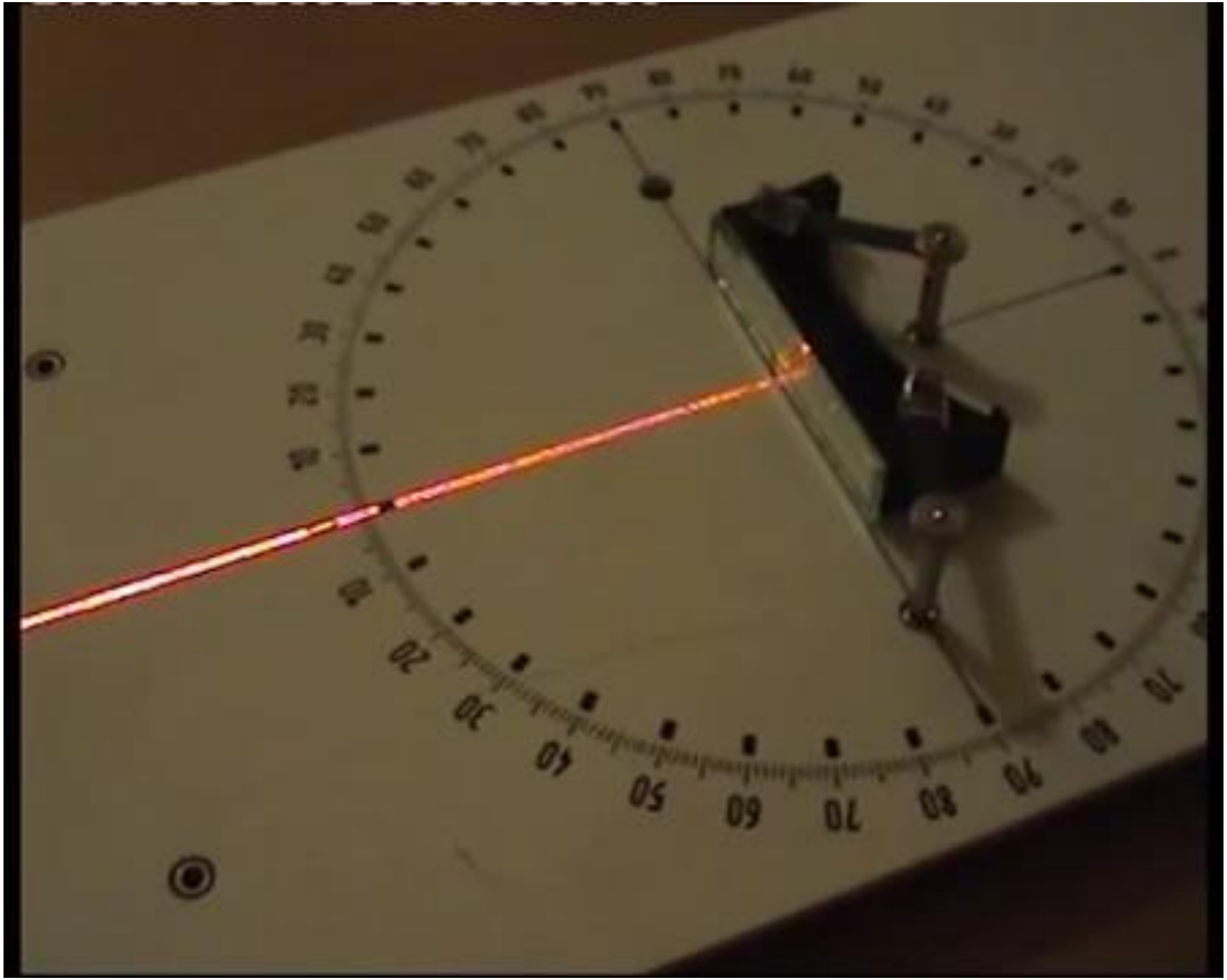


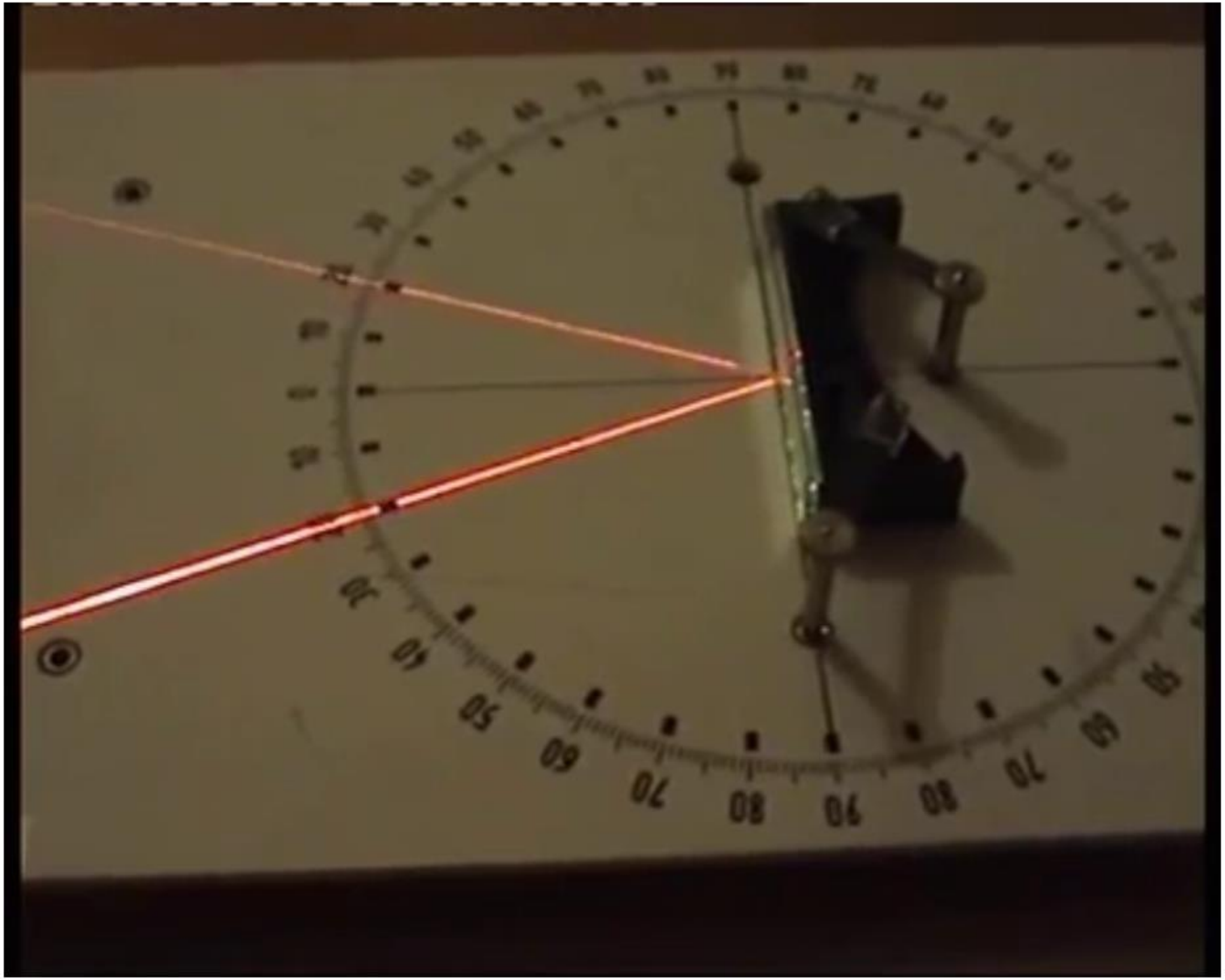


Second Law: The symmetry of reflected light is described by the 2nd law of reflection, which states that the angles of incidence and reflection are equal:

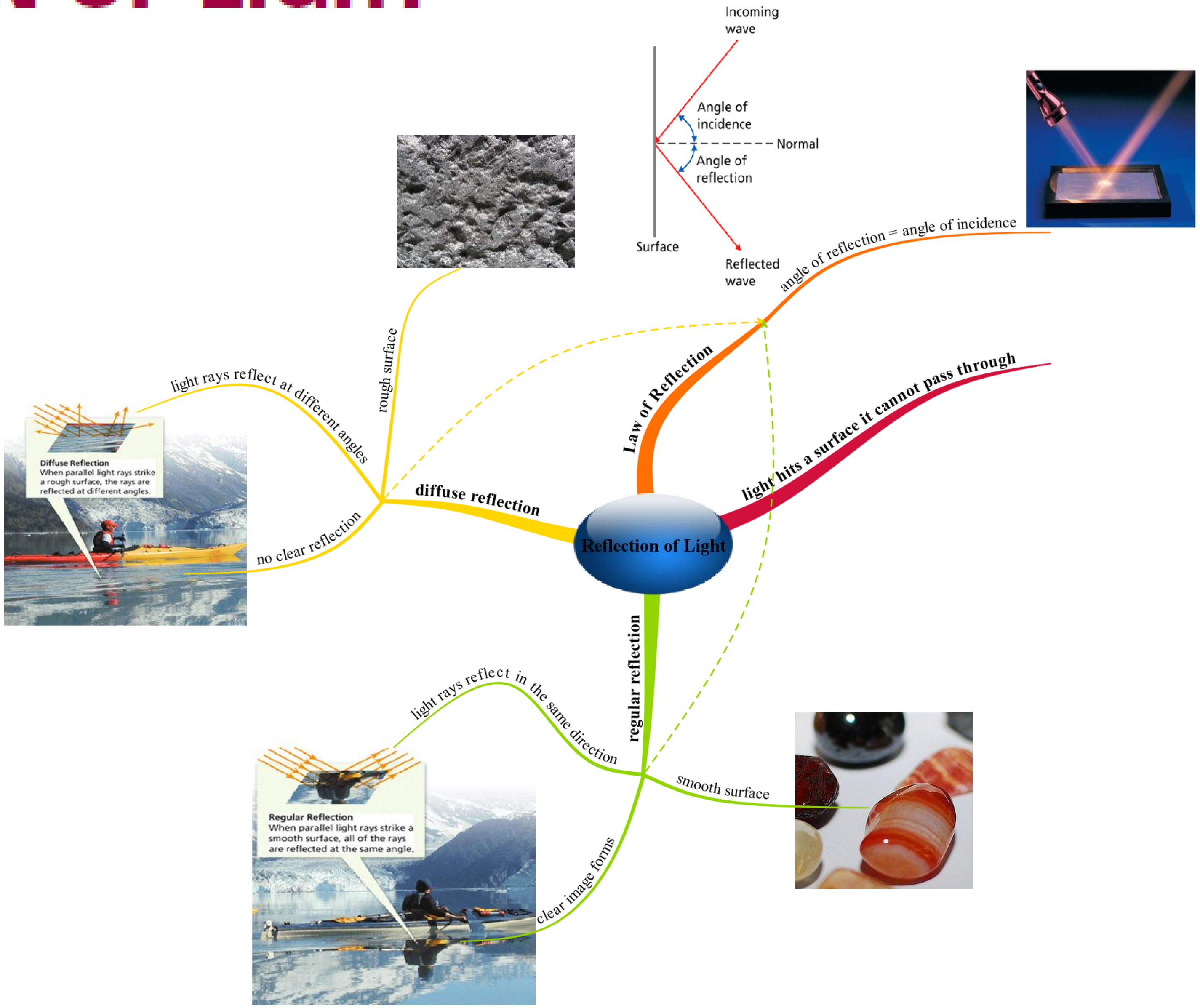
$$\boxed{\hat{\imath} = \hat{r}}$$

$$\boxed{\text{angle of incoming light ray} = \text{angle of reflected light ray}}$$



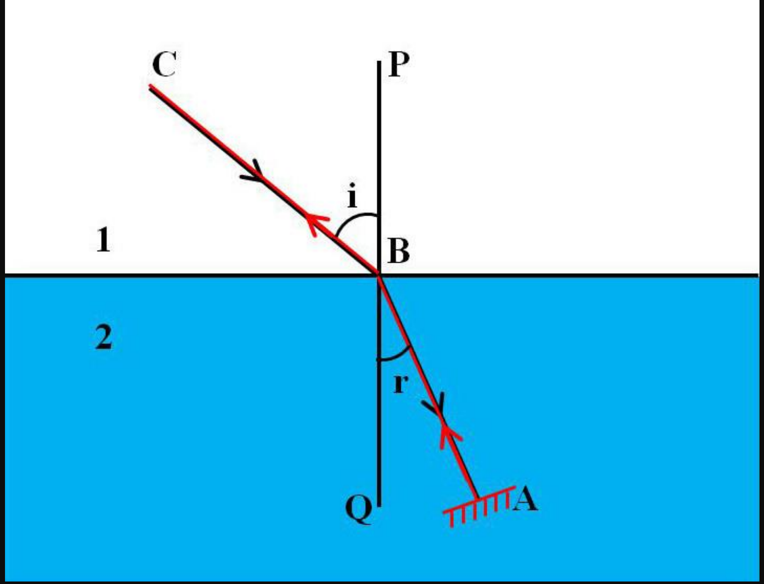


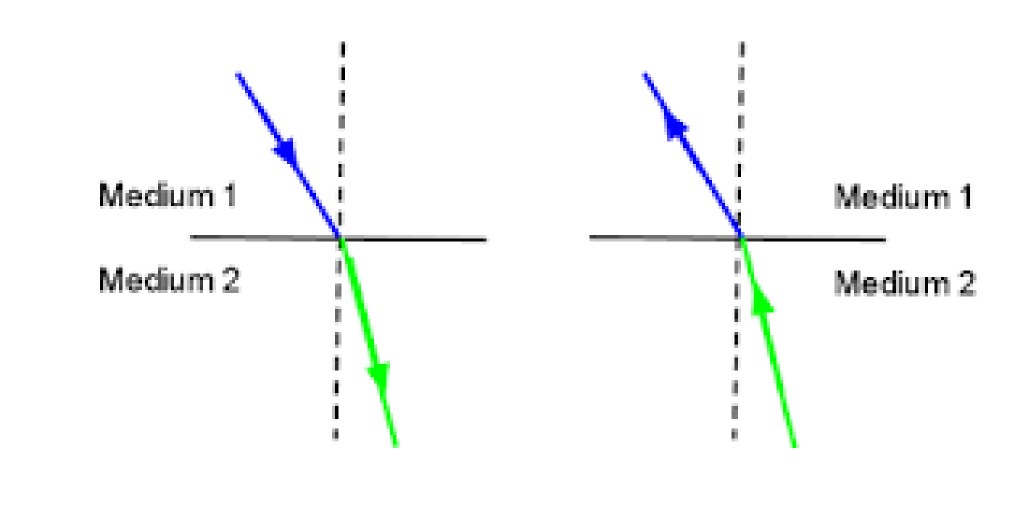
REFLECTION OF LIGHT

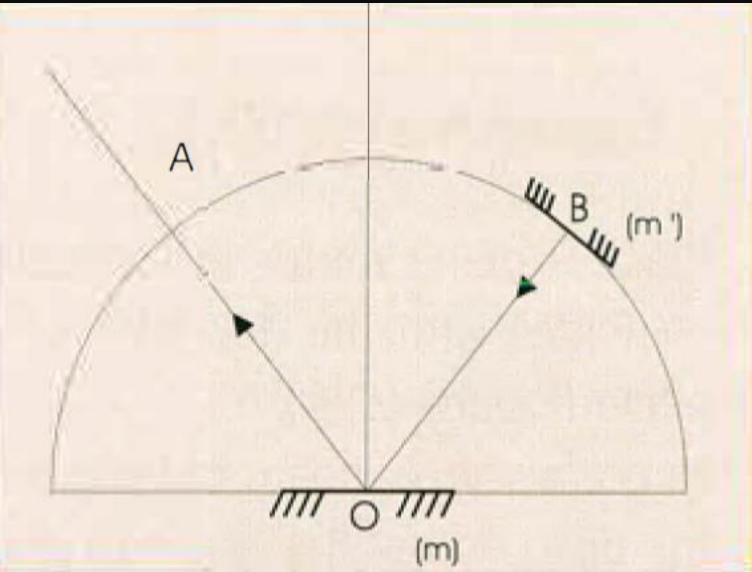


Reversibility of Light

The principle of reversibility states that light will follow exactly the same path if its direction of travel is reversed: *the path followed by light is independent of its direction of travel*.







Plane Mirrors

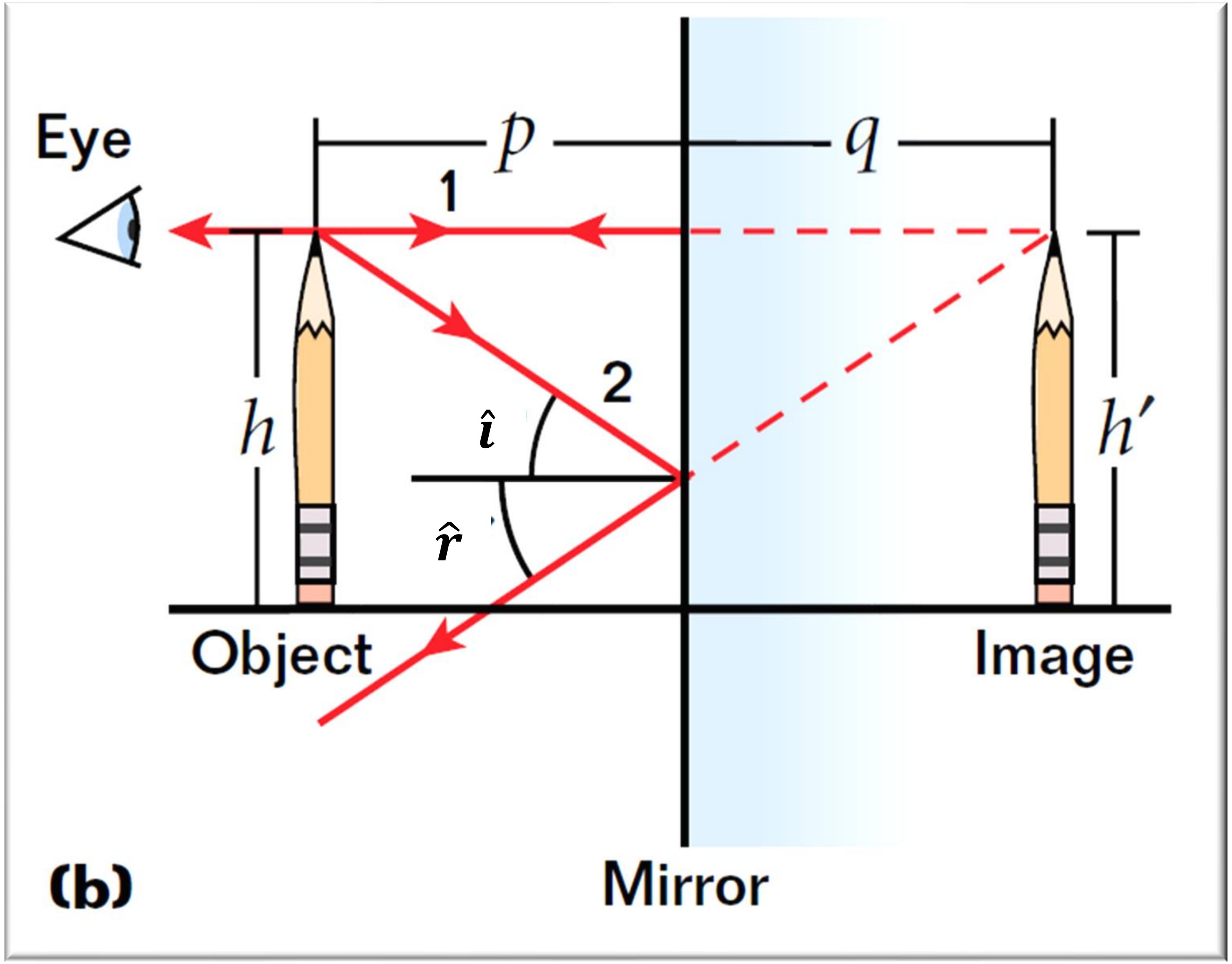
Plane Mirrors

- A plane mirror is a polished plane surface that reflects light falling on it.
- The most common plane mirrors are made of thin glass plates covered on one side with a silver or tin compound protected by paint.
- Other examples of plane mirrors are smooth plane metallic surfaces and still water surfaces.


reflection

Image location can be predicted with ray diagrams

Ray Diagram

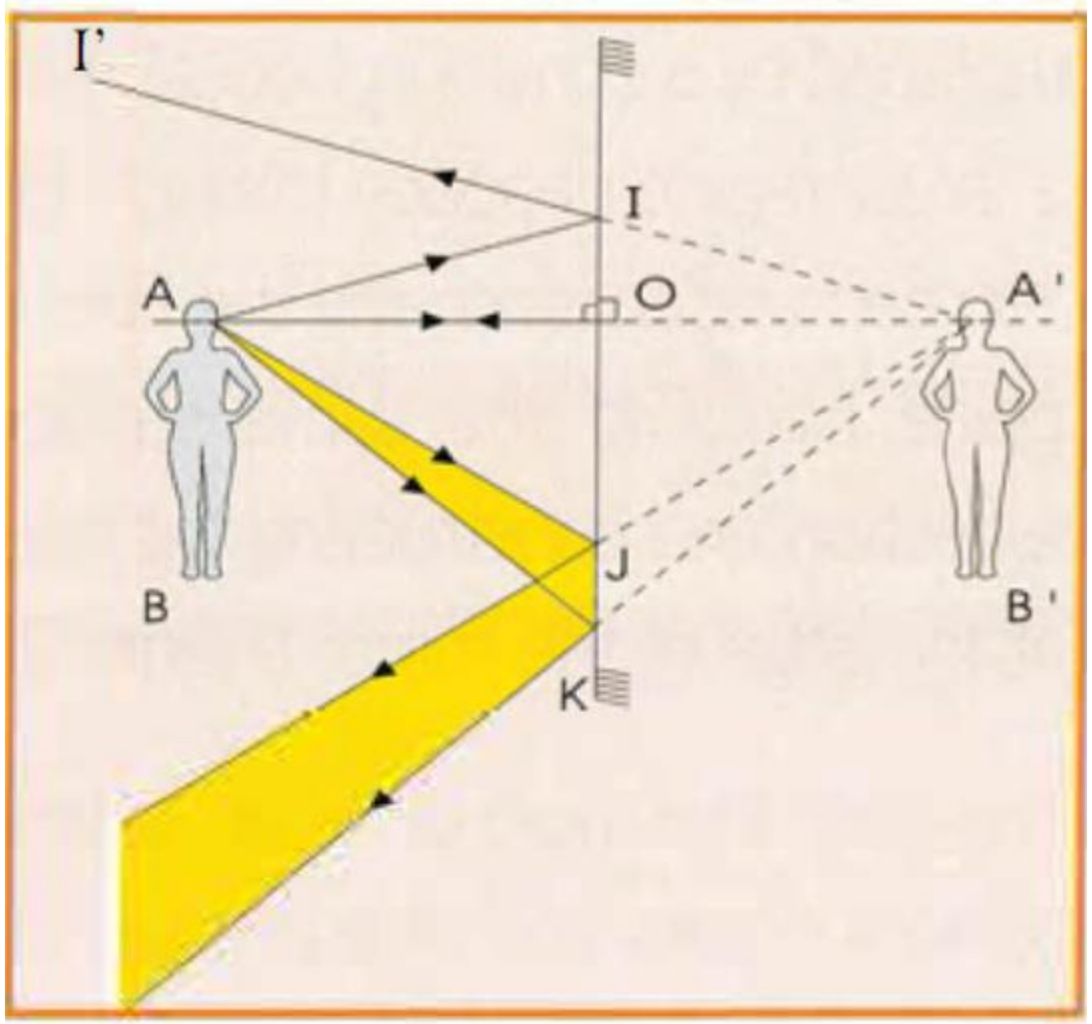


Construction of the image – Ray diagram(steps)

Consider a point object, represented by point A, placed in front of a plane mirror (M). To construct its image, consider two rays issued from A and incident on the mirror (Figure):

- A ray AO normal to the mirror and reflecting on itself.
- Any other incident ray AI reflecting along I I' such that i = r, (2nd law of reflection) . As A is the intersection of the two incident rays AO and AI, its image A' is the intersection of the corresponding reflected rays OA and I I'. These two rays do not meet in front of the mirror, but their backward extensions meet at A' behind M. A' is thus the virtual image of A seen through the mirror.

Note that any incident ray of light issued from A is reflected as if issued from its image A'.

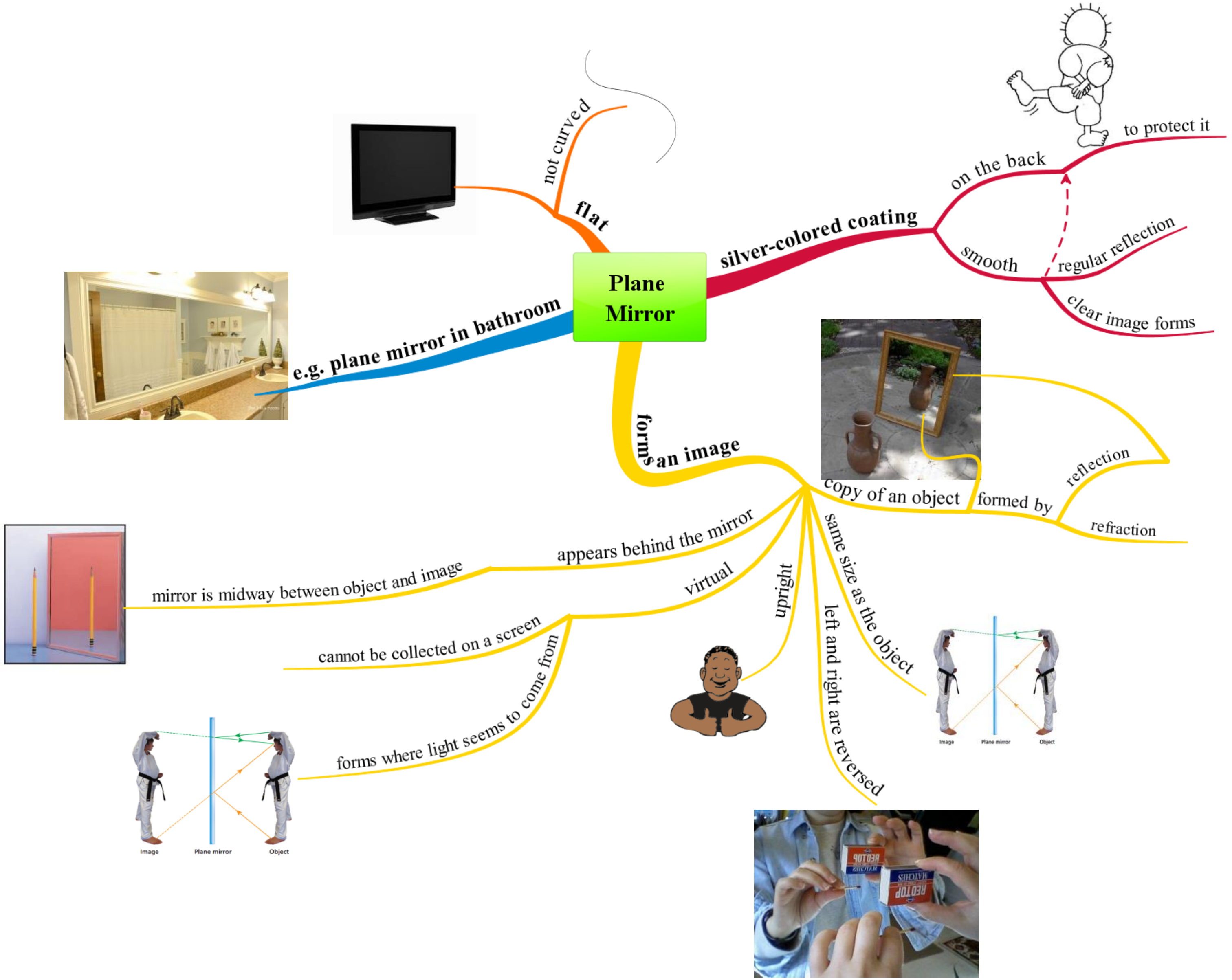


Conclusion: Characteristics of image given by a plane mirror

If an object is placed in front of a plane mirror, its image has the following properties:

- It is virtual.
- It is of the same size as the object.
- Its distance from the mirror is equal to that of the object from the mirror.
- It is not inverted along the vertical direction (it’s upright).
- It is laterally inverted (the image of a right hand is a left hand).

FLAT MIRRORS

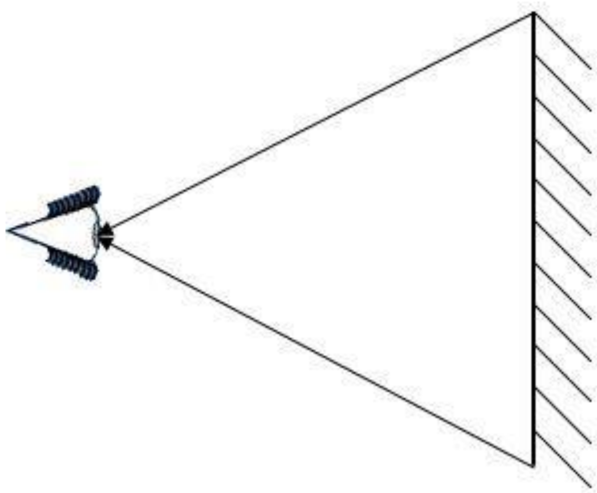


Field of Vision of a Plane Mirror - Definition

The **field of view** of a plane mirror is the space that an observer can perceive when looking in the mirror.

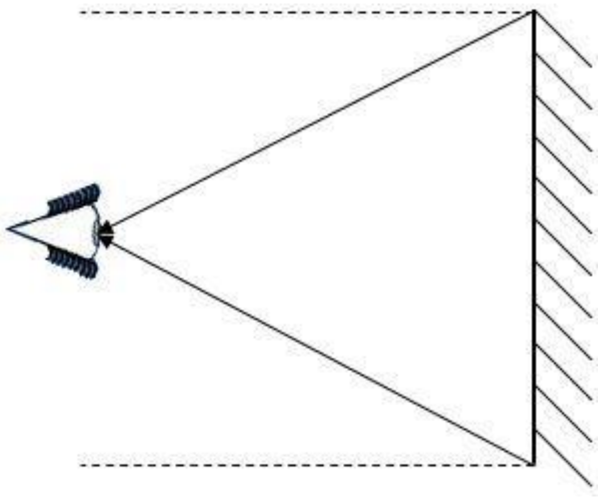
It is possible to determine the field of view using the laws of reflection. Simply follow the **steps** below to observe the field of view in a plane mirror.

1. Draw light rays from the ends of the plane mirror converging towards the eye of the observer.



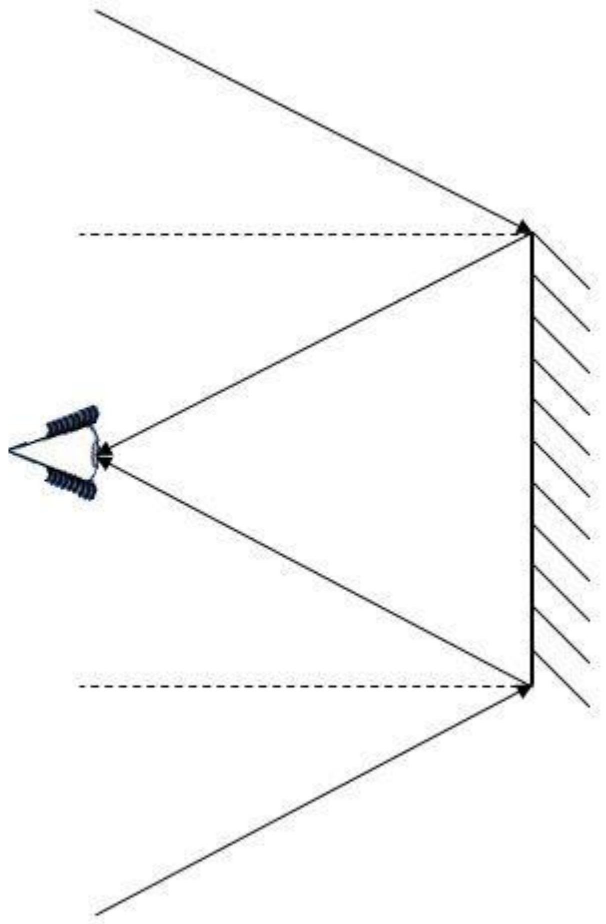
It is possible to determine the field of view using the laws of reflection. Simply follow the **steps** below to observe the field of view in a plane mirror.

2. Draw the normal at each point of incidence.



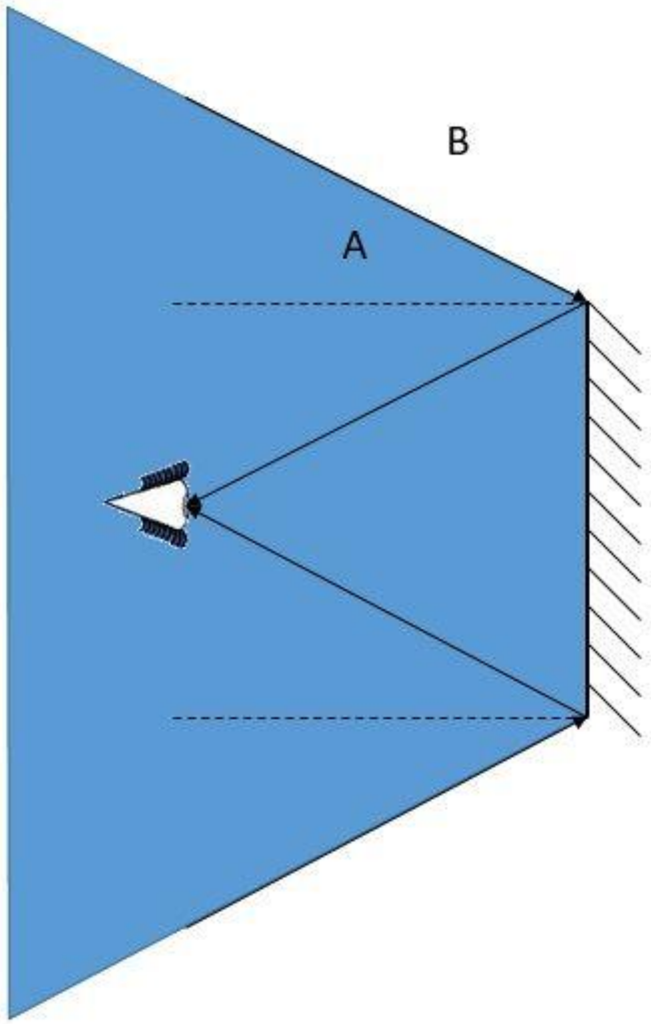
It is possible to determine the field of view using the laws of reflection. Simply follow the **steps** below to observe the field of view in a plane mirror.

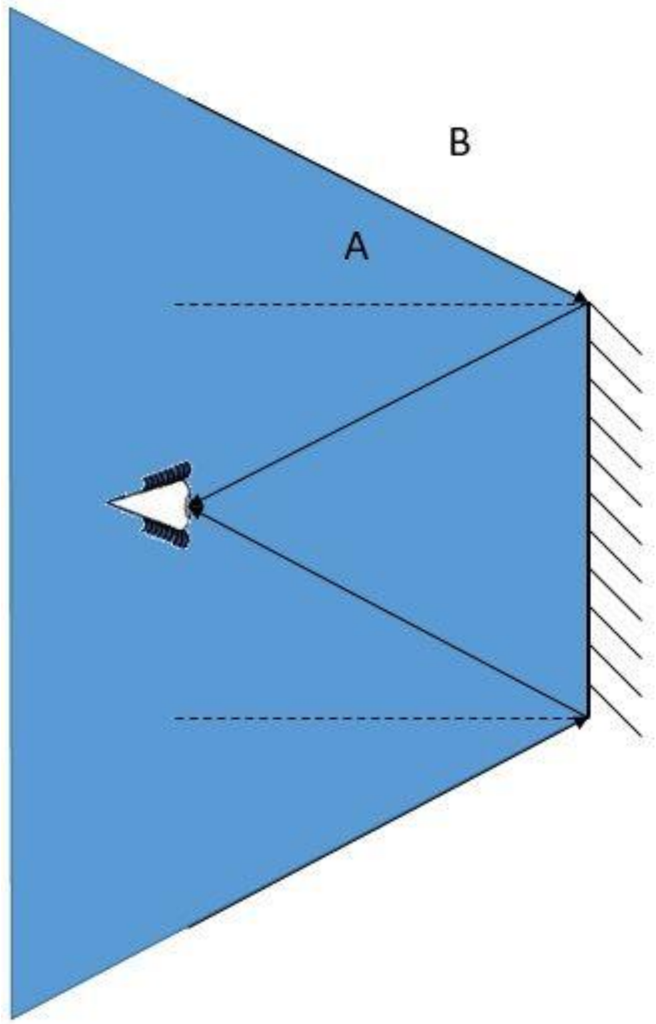
3. Draw the incident rays while observing the law of reflection.



It is possible to determine the field of view using the laws of reflection. Simply follow the **steps** below to observe the field of view in a plane mirror.

4. Everything between the incident rays is part of the observer's field of view.





Based on the illustration, it is possible to deduce that object **A would be within the observer's field of view**, while **object B would be outside his field of view**.

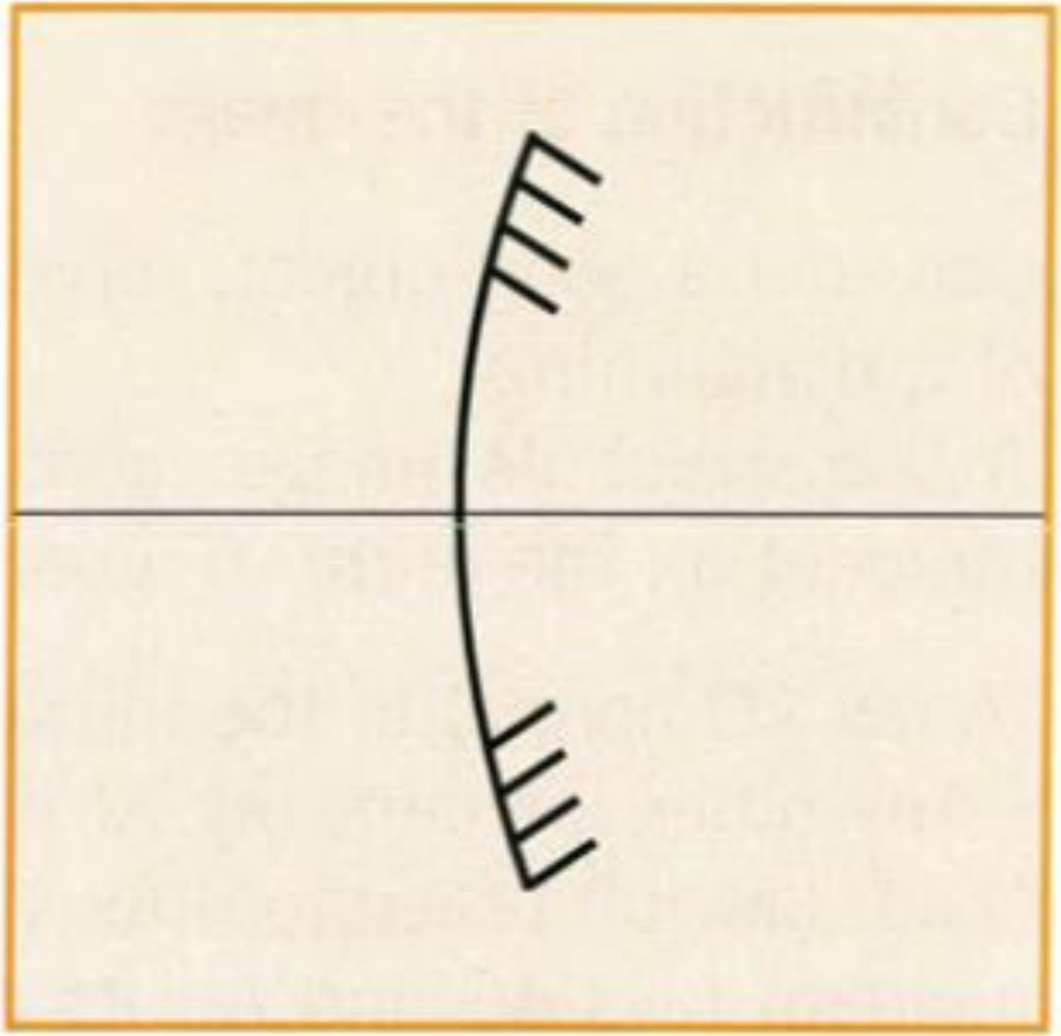
This technique can be used regardless of the viewer's position, whether in front of or next to the mirror.

To increase the field of vision in a plane mirror, it is possible to modify certain parameters:

- increase the size of the mirror. By choosing a larger mirror, the angles of incidence and reflection will increase, thus widening the field of view;
- bring the observer closer to the mirror. The angles of incidence and reflection will also increase, which will expand the field of view;
- use a convex mirror.

Convex mirrors

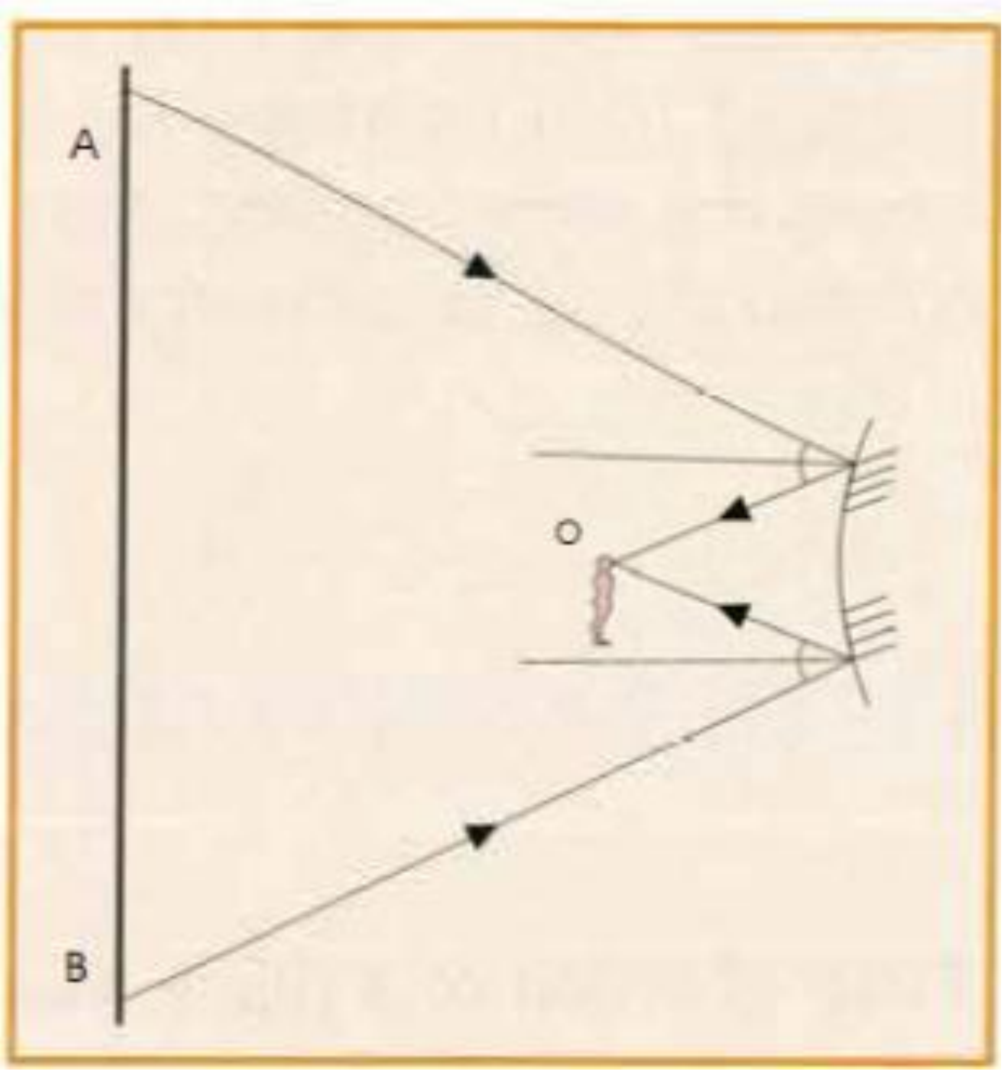
A convex mirror may have a spherical (left figure), cylindrical, or parabolic shape. It is usually made of glass for which the inner side is silvered.



Convex spherical mirror.

Field of vision of a convex mirror

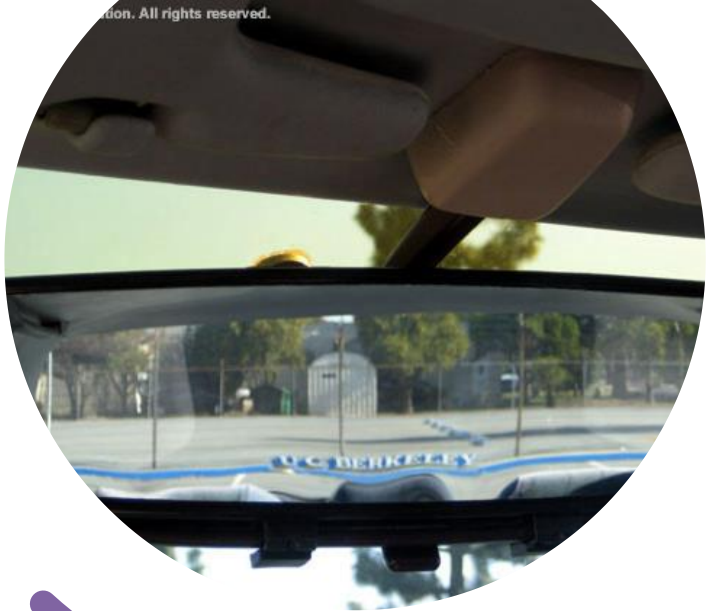
- For the same position of the observer's eye, if a circular plane mirror is replaced by a convex spherical mirror of the same size, the observer will see a larger region of space through that mirror.
- You can understand why the field of a convex mirror is larger than that of a plane mirror of the same surface area by applying the second law of reflection and constructing the rays that reach the eye after being reflected at the edge of the mirror



Field of vision of a convex mirror.

Field of vision of a convex mirror

Because their field of vision is larger, convex mirrors are used as rear-view mirrors in cars and motorcycles, and at crossroads.



Summary

* Reflection is the sudden change in the direction of propagation of light falling on a polished surface.
* When an incident ray falls on a reflecting surface, it is reflected and obeys the following laws:
  - The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane.
  - The angle of incidence is equal to the angle of reflection.
* The path of a ray of light is independent of its direction of propagation: this is the principle of reversibility of light.
* The image of an object, given by a plane mirror, is virtual and symmetric of the object with respect to the mirror.
* For the same position of an observer, the field of vision of a convex mirror is larger than that of a plane mirror of the same dimensions.

SECTION REVIEW

1. Which of the following are examples of specular reflection, and which are examples of diffuse reflection?
   - **a.** reflection of light from the surface of a lake on a calm day
   - **b.** reflection of light from a plastic trash bag
   - **c.** reflection of light from the lens of eyeglasses
   - **d.** reflection of light from a carpet

2. Suppose you are holding a flat mirror and standing at the center of a giant clock face built into the floor. Someone standing at 12 o'clock shines a beam of light toward you, and you want to use the mirror to reflect the beam toward an observer standing at 5 o'clock. What should the angle of incidence be to achieve this? What should the angle of reflection be?

3. Some department-store windows are slanted inward at the bottom. This is to decrease the glare from brightly illuminated buildings across the street, which would make it difficult for shoppers to see the display inside and near the bottom of the window. Sketch a light ray reflecting from such a window to show how this technique works.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4. **Interpreting Graphics** The photograph in **Figure 5** shows multiple images that were created by multiple reflections between two flat mirrors. What conclusion can you make about the relative orientation of the mirrors? Explain your answer.

5. **Critical Thinking** If one wall of a room consists of a large flat mirror, how much larger will the room appear to be? Explain your answer.

6. **Critical Thinking** Why does a flat mirror appear to reverse the person looking into a mirror left to right, but not up and down?